HWA CHONG INSTITUTION
JC2 Preliminary Examination
Higher 2

CANDIDATE NAME		**CT GROUP**	
CENTRE NUMBER		**INDEX NUMBER**	

GEOGRAPHY

9173/02

Paper 2

17 September 2025

3 hours

Additional Materials: Insert

READ THESE INSTRUCTIONS FIRST

An answer booklet will be provided with this question paper. You should follow the instructions on the front cover of the answer booklet. If you need additional answer paper ask the invigilator for a continuation booklet.

Section A.
Answer **Question 1**.

Section B.
Answer **Question 2**.

Section C.
Answer **one** question.

The number of marks is given in brackets [ ] at the end of each question or part question.

Answer Questions 1 and 2 and
EITHER Question 3 **OR** Question 4

Section A

Cluster 4 Fieldwork

1 A group of students from Singapore decided to investigate the influence of land use on infiltration rates. They did this to better understand fluvial flood risk in their neighbourhood. Having read literature on the subject, they suggested the hypothesis:

Residential areas reduce infiltration rates.

The students identified two sites for their investigation. Site X is next to a private residential high-rise building (Resource 1A), while Site Y is located within a nature park (Resource 1B).

They decided to collect the data on a weekday in August at 11am. The equipment the students used for their investigation was:

- A tin can, 12cm long, with both ends removed with a can opener
- A ruler and stopwatch
- Recording sheets (Resources 3A and 3B) and pencil.

The students used a hammer to sink the can into the ground. However, they only managed to do so to a depth of 1cm (Resource 1C).

Water was poured into the can until it was full. After a few minutes, the students recorded the drop in water level using a ruler. They then added water to bring the level back to approximately the original level at the start. They took approximately 30 minutes to collect data at Site X. Thereafter, they walked to Site Y (15 minutes' walk) and collected data for another approximately 30 minutes.

Resource 1A shows Site X next to a walled housing estate. Resource 1B shows Site Y situated within a nature park. Resource 1C shows the method used to measure infiltration rates at Site X. Resource 2 is a map of the study area where the students conducted the investigation. Resources 3A and 3B show the completed data collection sheets used during the investigation.

(a) Explain one strength and one weakness of the students' hypothesis. **[4]**

(b) Explain why the area shown in Resources 1A, 1B and 2, are suitable for the students to conduct this geographical investigation. **[5]**

(c) Explain precautions the students should take when conducting fieldwork at the sites shown in Resources 1A and 1B. **[6]**

(d) Explain the strengths and weaknesses of the data collection method shown in Resource 1C. **[5]**

(e) Sketch line graphs to represent cumulative infiltration (mm) over cumulative time (mins) shown in Resources 3A and 3B. **[6]**

(f) Explain why a line graph is an appropriate method to represent the data from Resources 3A and 3B. **[4]**

(g) Evaluate the students' investigation about infiltration rates affecting flood risk. **[10]**

[Turn over

Section B

Cluster 3 Sustainable Future and Climate Change

2 Resource 4 shows share of urban population in total (%) in Africa. Resource 5 shows the population pyramid of Ethiopia in 2021. Resource 6A shows the Setegn Meda slum in Addis Ababa, Ethiopia. Resource 6B shows the Integrated Housing Development Programme (IHDP) housing – Jemo condominium - in Addis Ababa, Ethiopia. Resource 7 shows perceptions of residents on comfort and positive characteristics of IDHP housing in three cities in Ethiopia.

(a) Cite data from Resource 4 to describe the changes in urban population across Africa between 1950 and 2010. **[4]**

(b) Suggest how population characteristics shown in Resource 5 may influence urbanisation rates in Ethiopia. **[6]**

(c) Compare the conditions of the two housing areas in Addis Ababa, Ethiopia shown in Resources 6A and 6B. **[4]**

(d) Explain the impact of multiple deprivations experienced by slum dwellers living in Addis Ababa, Ethiopia on sustainable urban development. **[4]**

(e) With reference to Resources 6B and 7, suggest why the Integrated Housing Development Programme (IHDP) in Ethiopia may have limited success. **[6]**

(f) Explain the ecological footprint of large cities in less developed countries. **[6]**

[Turn over

Section C

Answer EITHER question 3 OR question 4

Cluster 3 Sustainable Future and Climate Change

3 'Achieving a high level of liveability in cities for women is fraught with issues.

Evaluate this statement. **[20]**

4 Evaluate the extent to which impacts of climate change on people and places are similar. **[20]**

Acknowledgements
Question 1 Resource 1A,B,C *@ Author's own resource*
Question 1 Resource 2 *@ Google maps (Adapted)*
Question 1 Resource 3A,B *@ Author's own resource*
Question 2 Resource 4 *@ https://www.brookings.edu/wp-content/uploads/2020/07/200716_global_fotw_fig1.png*
Question 2 Resource 5 *@ https://www.cia.gov/the-world-factbook/*
Question 2 Resource 6A *@ https://www.habitatforhumanity.org.uk/blog/2017/11/slum-housing-ethiopia-modernise/*
Question 2 Resource 6B *@ https://www.reuters.com/article/world/ethiopians/*
Question 2 Resource 7 *@ https://www.thebritishacademy.ac.uk/documents/3530/JBA-9s9-04-SunikkaBlank-Abdie-Bardhan.pdf*